Gradient Descent Step-by-Step

$\alpha = 0.100$ is the learning rate, $x = 1.50$ is the input feature, and $y = 3.00$ is the target value.

$$Error = y - prediction = 0.11$$

$$L = (y - w_t \cdot x)^2$$

3.00 Target Value y | 1.925 Current Weight | 1.50 Input Feature x

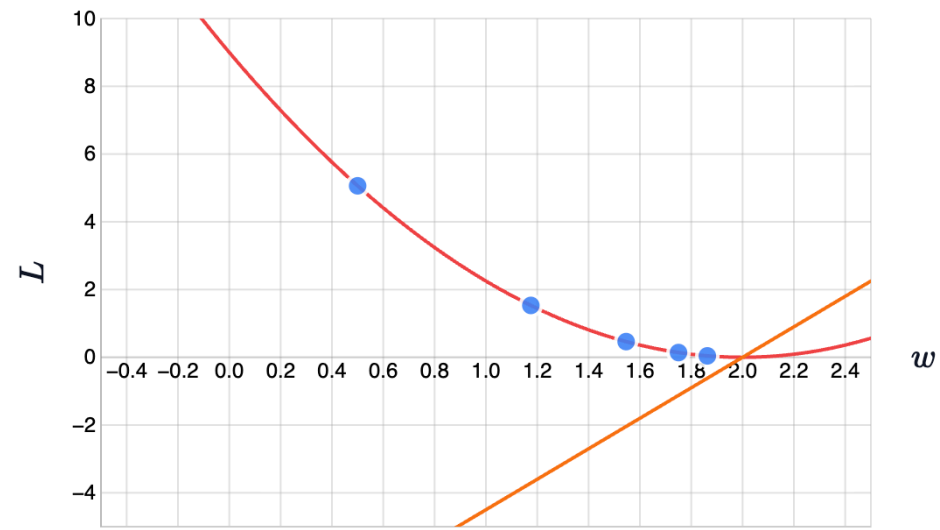
Calculating gradient

$$-2 \cdot 1.50 \cdot 0.11 = -0.34$$

$$\nabla L = -2x(y - w_t \cdot x)$$

-0.3397 Gradient

$$w_{t+1} = w_t - \alpha \cdot \nabla L$$



Loss L (parabola) — Gradient ∇L (line)

28 / 31